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30 September 1986
SUPERSEDING
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17 September 1969

MILITARY SPECIFICATION

GRINDING MACHINES, WOOD-CUTTING, CIRCULAR SAW BLADE, AUTOMATIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven, floor mounted, automatic, wood-cutting circular saw blade grinding machines with equipment and accessories specified herein.

1.2 Classification. The grinding machines shall be of the following sizes. The size to be furnished shall be as specified (see 6.2.1).

Sizes	Saw blade diameter range (inches)
1	6 to 24
2	20 to 72
3	20 to 84
4	20 to 96
5	20 to 108

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE REFERENCE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NFPA 70 - National Electrical Code.

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

ANSI B7.1 - Safety Requirements for the Use, Care and Protection of Abrasive Wheels.

ANSI B11.9 - Safety Requirements for the Construction, Care, and Use of Grinding Machines.

ANSI B74.2 - Specifications for Shapes and Sizes of Grinding Wheels and Shapes, Sizes, and Identification of Mounted Wheels.

ANSI B74.13 - Marking for Identifying Grinding Wheels and Other Bonded Abrasives.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws or regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of reproducing cross-cut and rip type wood-cutting circular saw tooth shapes utilizing automatic grinding and blade indexing cycles. The machine shall be capable of grinding saw blades of the geometry specified by the procuring activity (see 6.2.1). Means shall be provided to manually set up the machine and control saw blade and grinding wheel contact. The machine shall automatically grind all teeth on a saw blade after machine set-up by pushing one button. The machine shall be capable of maintaining uniform hook and tooth angles or restoring saw teeth points on saw blades within the saw blade diameter capacity of the machine. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators,

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pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station. The machine shall be automatic, allowing grinding of the saw blade in one set-up, in one operation, by pushing one button. Machine set-up shall be manually controlled by the operator.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with ANSI B11.9 and with all of the requirements of OSHA 29 CFR that are applicable to the machine itself. In the event of a conflict between the requirements of the ANSI and OSHA standards, the OSHA standards shall apply. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be so designed and constructed that under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation (see 6.4).

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3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger than 0.01 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position.

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3.3.8 Slides. All sliding members shall be constructed to meet and maintain the accuracy and alignment tolerances given in table II.

3.3.9 Gears. All gears and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears shall be of proper width and size to transmit full rated torque and horsepower throughout given speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell C-45.

3.3.10 Grinding wheels. Unless otherwise specified (see 6.2.1), grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1. All grinding wheels shall be provided with guards.

3.3.11 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. Each machine shall consist primarily of a base, grinding wheelhead assembly, tooth shaping mechanism, saw blade holder or carriage, drive components, and controls. Any additional components necessary for the operation of the machine shall be furnished in accordance with manufacturer's standard practice.

3.4.1 Base. The base shall be a box-like construction of cast iron or steel alloy, ribbed and braced to withstand forces imposed when the machine is utilized to its maximum capacities. The base shall support the grinding wheelhead assembly, the saw blade holder or carriage, the drive mechanisms and controls, and other related machine components. The base shall house the tooth shape generating mechanism and driving components. An access door(s) or removable plate(s) shall be provided for servicing components housed within the base. Means for floor mounting and leveling the machine shall be provided. When necessary, a sub-base shall be provided to allow grinding of large saw blades.

3.4.2 Grinding wheelhead assembly. The grinding wheelhead assembly shall be mounted to the base, composed of the wheelhead, spindle, drive mechanism, and related components. Means shall be provided to adjust the assembly vertically and horizontally to move the wheelhead over the saw blade to be ground for set-up. The assembly shall be capable of grinding saw teeth with an alternate bevel angle range between 0 and 45 degrees. The grinding wheelhead assembly shall automatically oscillate to grind face (hook) angles and gullets. The grinding wheelhead assembly shall be coordinated with the tooth shape generating mechanism (see 3.4.4) to reproduce required saw blade shapes.

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3.4.2.1 Grinding wheelhead. The wheelhead shall be properly guarded and house the grinding wheel spindle and related components. The wheelhead shall be coordinated with the tooth shape generating mechanism so the action of that mechanism will cause the proper saw blade shape to be ground. The wheelhead spindle shall rotate in the wheelhead assembly on bearings and accept a driving mechanism at one end and a grinding wheel at the other end. The wheelhead driving mechanism shall be powered by an electric motor. Unless otherwise specified (see 6.2.1), the drive shall be through a belt and pulley(s). Means for adjusting belt tension shall be provided unless a nontensioning type belt is used. Grinding wheel characteristics, speeds and motor requirements shall be in accordance with table I.

3.4.3 Saw blade holder or carriage. The saw blade holder or carriage shall be mounted to the side of the base and shall accept saw blades within the diameter range for the size machine specified. Means for mounting and holding the saw blade on the holder or carriage shall be provided. The holder or carriage shall be vertically and horizontally adjustable to accommodate saw blades of various diameters. The saw blade mounting device shall permit 360° indexing of the saw blade. An adjustable clamp or support shall be provided to prevent saw blade vibration during grinding.

3.4.4 Tooth shape generating mechanism. The tooth shape generating mechanism shall be housed in the base and be easily accessible. The mechanism shall reproduce required tooth shapes through the grinding wheelhead assembly to the saw blade. The arrangement shall include means for interchangeability to provide for a series of tooth shapes. The tooth shape generating mechanism shall be adjustable for various tooth spacings and depths. Dials or scales shall indicate tooth hook and bevel angles. Unless otherwise specified (see 6.2.1), a variable lift feature shall be incorporated in the feed system. Means shall be provided, through the variable lift, to adjust the effect of the lift cam to allow either deep or shallow teeth to be sharpened without changing cams. Unless otherwise specified (see 6.2.1), the tooth shape generating mechanism shall be powered by an electric motor, independent from the grinding wheelhead assembly, and shall be driven through a belt and pulley(s). Means for adjusting belt tension shall be provided unless a nontensioning type belt is used. Unless otherwise specified (see 6.2.1), the tooth shape generating mechanism and its feed system shall meet the motor horsepower, speed range, and sharpened tooth production rate given in table I.

3.4.5 Dust extractor. When specified (see 6.2.1), a dust extractor to be utilized during dry grinding shall be provided. The dust extractor shall be connected to the machine's electrical system with start/stop pushbuttons conveniently located on the control panel.

3.4.6 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit. Each machine shall have a fused safety disconnect switch or

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circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine upon installation shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.6.1 Motors. Motors shall be rated for continuous duty, shall be dual voltage 230/460-volt high efficiency type, and shall have ball or roller bearings of the sealed and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 60-Hz. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall meet the requirements of ANSI/NEMA MG 1 and ANSI/NFPA 70.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than specified in table I for the machine size specified.

3.6 Performance. The machine shall be capable of grinding saw blades in accordance with the requirements herein and specifically in accordance with the test requirements designated in 4.7.

3.7 Alignment accuracy requirements. Each machine shall meet the alignment and accuracy tolerances specified in table II when tested according to the procedures given in 4.7.4.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of special hand tools normally furnished by the manufacturer.

b. One worklight, adjustable to illuminate the work area, complete with protective shield, on-off switch, and mounting hardware.

c. Tapered cones or bushings in increments not to exceed 1/8 inch for saw blades with arbor holes for the range given in table I for the size machine specified.

d. Tooth shape generating cams, the number of each, type, style and all other essential information as designated by the procuring activity (see 6.2.1).

e. One extra set of wheelhead drive belts.

f. One extra set of tooth shape generating mechanism drive belts, when applicable (see 3.4.4).

g. Six grinding wheels, type, dimensional characteristics and composition to be as specified (see 6.2.1).

h. Wheel dresser, either permanently mounted or independent (see 6.2.1).

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3.9 Optional equipment. Optional equipment and accessories shall be furnished as specified (see 6.2.1).

3.10 Repair parts. Such repair parts as are specified (see 6.2.1) shall be furnished.

3.11 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion-resistant.

3.11.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix indentified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Technical data. When technical data (operating manuals, maintenance manuals, part catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the machine shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own

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or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under normal environmental conditions.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6, all tests in 4.7 and the inspection in 4.8. Failure of the item to pass the first article examination or all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, or inspection shall be cause for rejection.

4.6 Examination. The machine and equipment shall be examined for compliance with the requirements in 3.2 through 3.5 and 3.8 through 3.13, including compliance with all referenced standards.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than 1 hour. Proper operation of all controls, motors, adjusting mechanisms, and other accessories shall be verified during the trial period.

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4.7.2 Performance test. Each machine shall be subjected to a grinding test. Saw blade(s) shall be ground in accordance with the requirements specified by the procuring activity (see 6.2.1) including the number of blades, range in gage, range in diameters, tooth spacing from point to point, tooth hook and bevel angles, tooth style(s), diameter of saw eyes, acceptability criteria, and other necessary requirements for testing machine performance. Unless otherwise specified (see 6.2.1), the procuring activity will furnish the saw blade(s) to be ground. The method used to perform the test shall be at the supplier's option provided it accurately determines the capability of the machine.

4.7.3 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.11.

4.7.4 Alignment accuracy test. Each machine and components shall be checked for compliance with the accuracies specified in table II. The instruments and methods used in performing the tests shall be at the option of the supplier provided they accurately determine the tolerances of the machine.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The grinding machines covered by this specification are intended for use in any woodworking shop where circular saw blades are used and need to be resharpened.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. First article, if required (see 3.1).

d. Geometry of saw blade(s) to be ground (diameter, tooth spacing, tooth depth, tooth hook and bevel angles, tooth style and saw eye diameter) (see 1.2).

e. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

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f. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. Grinding wheels, if different (see 3.3.10).

j. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.11).

k. Grinding wheelhead drive, if different (see 3.4.2.1).

l. Variable lift, if not required (see 3.4.4).

m. Tooth shape generating mechanism motor and drive requirements, if different (see 3.4.4).

n. Tooth shape generating mechanism characteristics, if different (see 3.4.4).

o. Dust extraction unit, if required (see 3.4.5).

p. Electrical system and input power, if different (see 3.4.6).

q. Motor enclosure, if different (see 3.4.6.1).

r. Size and capacity, if different (see 3.5)

s. Standard equipment, if different (see 3.8).

t. Number, type, style, and other information for tooth shape generating cams (see 3.8.(d)).

u. Grinding wheels, type, dimensional caracteristics, and composition required (see 3.8.(g)).

v. Wheel dresser, as specified (see 3.8(h)).

w. Specify and fully describe optional equipment, if required (see 3.9).

x. Specify applicable repair parts, if required (see 3.10).

y. Lubrication plate, if different (see 3.11.1).

z. Nameplate, if different (see 3.11.2).

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aa. Inspection conditions, if different (see 4.3).

bb. First article inspection, if different (see 4.4).

cc. Quality conformance inspection, if different (sse 4.5).

dd. Specify and fully describe performance test; saw blades to be ground to be furnished by manufacturer, if required (see 4.7.2).

ee. If preservation, packing, and marking in accordance with MIL-M-18058 are required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, part lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machine in the environment), the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional requirement that will integrate the new machine into its future operating environment.

6.5 Subject term (key word) listing.

Abrasive wheel
Circular saw blade, woodworking
Grinder, circular saw blade
Machine tool, grinding, circular saw blade
Saw blade, circular
Saw blade holder
Tooth shape generating mechanism
Wheel, abrasive, grinding
Wheelhead, grinding
Woodworking, saw blade

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6.6 Cross-reference of classification changes. Machine classification changes from the previous issue are as follows:

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Size 214	Deleted
Size 424	Deleted
Size 624	Size 1
Size 840	Size 1 and Size 2
Size 936	Size 1 and Size 2
Size 1160	Size 1 and Size 2
Size 1272	Size 1 and Size 2
Size 1360	Size 1 and Size 2
Size 2479	Size 2 and Size 3
Size 40118	Size 5
Size New	Size 4

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Sizes and capacities.

Characteristics	Requirements				
	Size 1	Size 2	Size 3	Size 4	Size 5
Saw blade diameter, range, inches	6 - 24	20 - 72	20 - 84	20 - 96	20 - 108
Feed, teeth per minute	← 25 →				
Tooth space, range, inches	1/4 - 2	← 3/8 - 3 →			
Tooth depth, range, inches	1/4 - 1	1/4 - 2	← 1/4 - 3 →		
Tooth hook, range, degrees					
Positive	0 - 40	0 - 40	0 - 40	0 - 40	0 - 40
Negative	0 - 20	0 - 20	0 - 20	0 - 20	0 - 20
Tooth face bevel, range, degrees	0 - 45	0 - 45	0 - 45	0 - 45	0 - 45
Saw hole, range, inches	1/2 - 2	3/4 - 9	3/4 - 12	3/4 - 15	3/4 - 15
Grinding wheel speed, RPM	← 1750 →				
Surface feet per minute, SFPM	← 5500 →				
Grinding wheel, inches					
Diameter	8	12	12	12	12
Thickness	To Suit	To Suit	To Suit	To Suit	To Suit
Bore	3/4	← 1-1/4 →			
Motors, horsepower					
Wheelhead drive	3/4	← 1 →			
Tooth shape generating mechanism	3/4	← 1/2 →			

NOTE: Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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TABLE II. Alignment and tolerance requirements.

Characteristic	Maximum allowable deviation
Tooth positioning accuracy	±0.003 inch
Vertical grinding wheelhead assembly repeatability	±0.002 inch
Head spindle runout	±0.001 inch T.I.R.
Grinding wheel spindle runout	±0.001 inch T.I.R.

Custodians:
Army - AL
Air Force - 99

Review activities:
Army - CR, ER, SM
Air Force - 84
DLA - GS

User activities:
Army - AR, ME, SM
Navy - MC, AS

Preparing activity:
DLA - IP

Project (3415-0159)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-80189A	Grinding Machines, Wood-cutting, Circular Saw Blade, Automatic

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 *PREVIOUS EDITION IS OBSOLETE.*

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